Volocopter 2X

We've been expecting flying cars to show up for a while now, despite the clear technical limitations of years gone by. This year, at CES 2018, we got to see not only a prototype, but a production version of what could be an actual autonomous air taxi in the near future. Yes, we are talking about the Volocopter. Let us be clear, it would be better to say that the 18-rotor Volocopter 2X is more a human-carrying drone than a flying car, but it is still cool! While rival NVIDIA might have the upper hand when it comes to autonomous vehicles on the ground, this could be Intel's way of reclaiming their stake in the game. Why is this significant? Because for a change, we get to see Intel CEO Brian Krzanich actually flying a Volocopter 2X during the Intel keynote. The 2X offers impressive numbers like a 27km range at 70kmph, or 27min at 50kmph and can go as fast as 100kmph. The design includes multiple redundancies in all critical components like rotors, power source and more. Promising days are ahead for flying taxis.
http://dgit.in/VoloDemo

Sony Aibo

Back in 1999, Sony introduced a robotic pet with Aibo, and up until 2005, released new models every year. However, 11 years after the original Aibo was discontinued, what Sony unveiled at CES 2018 makes the original looks like child's play. For starters, the upgraded Aibo has OLED eyes that make it capable of a more diverse and accurate range of expressions. It also packs AI capabilities, that let it develop a personality over time. The same capabilities also let it recognise people in your house and also interact with them based on who pets it the most. In terms of sensors, it has a camera on the nose and touch sensors on its head, chin and back. The robot has been rebuilt to be more dog-like, so that it can perform a range of actions like shake its head, find a ball or a bone, get a hi-five, or even lie down. Priced at $1,740, the bot needs to be connected to the internet via an LTE SIM to enable its learning capabilities, which are hosted on the cloud. It also needs a subscription of about $27 a month or $800 for three years. **http://dgit.in/AiboDemo**

Vivo in-display fingerprint sensor

A couple of years ago, dual-cameras and super-wide displays on smartphones would be unfeasible, whereas they've practically begun to flood the market at the moment. In the same vein, albeit a bit later, on-display fingerprint sensors have been a subject of great debate and speculation in the world of smartphone technology. While we've seen earlier demonstrations that have proposed piezo-based implementations, Vivo, the Chinese smartphone manufacturer, has actually displayed a smartphone this year that has a fully functional in-display fingerprint sensor. Powered by a Synaptics sensor that is (probably) the Clear ID FS9500, the implementation works perfectly as advertised and is most likely coming with the recently listed Vivo X20 Plus UD (with UD possibly meaning 'under display' for this particular phone).

The sensor is essentially a small CMOS device that goes below the AMOLED display and is able to peek through the OLED pixels at the reflected fingerprint. According to Synaptics, the sensor takes about 0.7 seconds to authenticate a fingerprint, which is still faster than the 1.4 seconds it takes for face-based authentication to work. The only difference at this point seems to be a more thorough registration process and the lack of haptic feed-

back. It would be interesting to see if long term pixel-burn in on OLEDs also affects the performance as well. But if you had any doubts that on-display fingerprint sensors will be coming to consumer smartphones this year, this should be sufficient to quell the same.

Who knows, we might even see on-display fingerprint sensors show up on a lot of other devices. For instance, smart-watches with OLED displays would benefit with something like this. As we have already seen, ambient light sensors can be placed under displays on smartwatches. If a similar design is implemented using the in-display fingerprint technology, that would bring an added level of independence to the smartwatch, making it more capable as a separate device that does not need to be paired with your smartphone. But at this point, it all boils down to how quickly the industry picks this up! **http://dgit.in/VivoFPS**